

SATHISH G

Software Engineer | Backend Engineer | AI Engineer | Python

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Summary

Software Engineer with 3+ years of experience designing and shipping production **GenAI** and backend systems, including two independently owned platforms now serving **10,000+ daily active users**. Hands-on experience building **RAG pipelines** (**LangChain**, **PGVector** hybrid semantic + BM25 retrieval), **agentic AI** workflows (**LangGraph**, **Claude Agent SDK**), and multi-tenant **RBAC** architectures, backed by strong fundamentals in **Python**, **SQL**, **RESTful API** design, and **microservices** across the full development lifecycle. Experienced in **Agile** delivery, **CI/CD automation** (cut deployment time by **73%**), and cloud infrastructure on **Azure** and **AWS** with **Docker/Kubernetes**. Proficient with AI-assisted development tools (**GitHub Copilot**, **Claude Code**) to accelerate coding, code review, and test generation.

Technical Skills

AI / GenAI: LangChain, LangGraph, Claude Agent SDK, RAG Pipelines, Hybrid Semantic + BM25 Retrieval, Azure OpenAI, Agentic AI, Loop Engineering, Prompt Engineering

Programming Languages: Python, Java, JavaScript, TypeScript

Frontend: Angular, HTML5, CSS3, Responsive Design, Chrome Extensions (Manifest V3)

Backend: REST APIs, Microservices, Node.js, FastAPI, Server-Sent Events (SSE)

Databases: PostgreSQL, SQL, MongoDB, Redis, PGVector, Elasticsearch, ADLS, S3

Cloud & DevOps: Microsoft Azure, AWS, Docker, Kubernetes (AKS, EKS), CI/CD, GitHub Actions

AI-Assisted Development: GitHub Copilot, Claude Code

Testing & Quality: Unit Testing, Integration Testing, Debugging, Code Review

Tools: Git, Postman, Jira, Azure DevOps

Concepts: Data Structures & Algorithms, System Design, Multi-Tenant Architecture, RBAC, Performance Optimization, Root Cause Analysis, Agile, SDLC

Professional Experience

Software Engineer

Medha Analytics AI, Bangalore, Karnataka

May 2025 – Present

Associate Software Engineer

Medha Analytics AI, Bangalore, Karnataka

Jul 2023 – May 2025

- Developed frontend UI components (**Angular**) and backend microservices (**Python**, **Node.js**, **FastAPI**) based on functional and technical requirements, following established coding standards.
- Designed and implemented **RESTful APIs** and integrated third-party services (WhatsApp Business API, LLM providers, Salesforce) across multiple production systems.
- Wrote and maintained unit/integration tests, collaborated with QA to resolve defects, and participated in code reviews and **Agile** ceremonies (stand-ups, sprint planning, retrospectives) to keep delivery on track.
- Used **GitHub Copilot** and **Claude Code** daily to accelerate development, code review, and test generation.
- Replaced manual, VM-based deployments with automated **CI/CD** pipelines (**GitHub Actions**), cutting deployment time **73%** (45 to 12 minutes), and deployed containerized workloads on **Kubernetes** (Azure AKS, AWS EKS).
- Diagnosed and resolved a large-dataset processing bottleneck, improving throughput by **35%**, and performed root-cause analysis on production incidents to restore service with minimal disruption.

Key Projects

YASA – Scalable Spreadsheet-Based Web Application

Angular, Node.js, Python, MongoDB, PostgreSQL

- Designed and built a full-stack, Excel-like data management platform enabling users to create, manage, and securely share structured data, with an **Angular** frontend and **Node.js/Python** backend microservices.
- Implemented granular **RBAC** (per-user view/edit permissions) and **RESTful APIs** across **MongoDB** and **PostgreSQL** for secure, multi-role data access.
- Diagnosed and resolved a large-dataset processing bottleneck, improving throughput by **35%**, and contributed to the system's scalable architecture and deployment workflows.

AiroStrip – AI-Powered Chrome Extension (Solo Build)

Python, FastAPI, PostgreSQL, PGVector, Angular, JavaScript (Chrome Extension, Manifest V3)

- Independently designed and built an **AI-powered Chrome extension** for Medha Analytics AI's healthcare products, bringing AI chat and dashboard access directly into the browser, driving a **30%** increase in user adoption.
- Developed backend APIs for dashboard access, AI chat, and chat history, backed by **PostgreSQL** with **RBAC** and a multi-tenant permission system enforcing per-tenant/per-user data isolation, with real-time cache invalidation for consistency.
- Built a retrieval pipeline using **PostgreSQL PGVector** with hybrid semantic + keyword (**BM25**) search for efficient querying across tenants.
- Built the Chrome extension (**Manifest V3**) frontend and companion **Angular** web app, integrating **Microsoft SSO**

authentication for secure access.

Cloud Deployment & Infrastructure Setup

Azure, AWS, Docker, Kubernetes, CI/CD, GitHub Actions

- Replaced manual, VM-based production deployments with fully automated **CI/CD** pipelines (**GitHub Actions**), eliminating manual intervention and cutting deployment time by **73%** (45 to 12 minutes).
- Deployed and orchestrated containerized workloads (**Docker, Kubernetes**) across both **Microsoft Azure (AKS)** and **AWS (EKS)**, managing environment configuration, DNS/SSL, and release processes end-to-end.
- Triaged production issues and performed root-cause analysis across deployed services, restoring service with minimal customer disruption.

Appointment Mitra – WhatsApp Booking Assistant (Solo Build)

Python, FastAPI, PostgreSQL, WhatsApp API, LangGraph, NLP, Agentic AI

- Independently designed and built a **WhatsApp Business API** chatbot automating hospital appointment booking, now handling **500+ bookings/month** across **1,500+ monthly active users**.
- Engineered the conversational engine combining a custom **NLP-driven** multi-turn dialogue state machine with an **agentic, loop-engineered** workflow built using **LangGraph**, with geolocation-based hospital matching.
- Integrated with backend and payment services for secure, real-time data handling across live patient booking flows.

Website AI Chatbot Widget – Live on narayanahealth.org

Python, FastAPI, LangChain, PostgreSQL/PGVector, JavaScript, Trino

- Designed and shipped an embeddable, plug-and-play **AI chatbot widget** that integrates into any webpage via a single script tag, now live on **www.narayanahealth.org**, serving **1,000+ daily active users** and driving **5,000+ appointment bookings/month**.
- Built a **RAG** pipeline with document ingestion (PDF, DOCX, PPTX) into **PostgreSQL PGVector** for context-aware responses, backed by a **FastAPI** backend and **Trino** for federated data queries.
- Designed a lightweight, configurable JavaScript widget (bundled with esbuild) with customizable branding and API integration for easy embedding across hospital web properties.

Education

Bachelor of Engineering, Computer Science

Government College of Engineering, Erode

2019 – 2023

CGPA: 8.18/10

Certifications

- [Software Engineer \(Problem Solving, SQL & REST API\) – HackerRank](#)
- [System Design: Low-Level to High-Level Solutions – GeeksforGeeks](#)
- Generative AI Applications Using Amazon Bedrock – AWS

Additional Information

Problem Solving: **300+** problems solved across LeetCode and HackerRank